

Coding and Smart Systems Outreach Program

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Traffic Light

Hardware Components Required:

- Raspberry Pi 4 with GPIO extension
- Breadboard
- 3 LEDs (Red, Yellow and Green)
- 3 Resistors (each 220 Ω)
- Wires: yellow, orange, green, black colors

Step-by-step instructions to connect hardware components:

1. If the LED already on the breadboard is not Red, replace it with a Red LED.
2. Plug **Yellow** LED next to Red LED on the breadboard so that long and short legs are in separate new empty rows.
3. Plug **Green** LED next to Yellow LED on the breadboard so that long and short legs are in separate new empty rows.
4. Connect one end of a 220 Ω resistor to the **long leg of Yellow LED** and the other end **to a new empty row** (resistor row).
5. Connect one end of a 220 Ω resistor to the **long leg of Green LED** and the other end **to a new empty row** (resistor row).
6. Connect the Yellow LED resistor row to GPIO 27 (pin 13) using a wire.
7. Connect the Green LED resistor row to GPIO 22 (pin 15) using a wire.
8. Connect all LED short legs to the GPIO GND (pin 34) using black wires.
9. Power on the Pi and click on the projects folder.
10. Click on the traffic_light.py file and write the code.

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Python code (Solution):

```
1 # traffic_light.py
2 from gpiozero import LED
3 from time import sleep
4
5 # Define LEDs
6 red_led = LED(17)
7 yellow_led = LED(27)
8 green_led = LED(22)
9
10 print("Traffic light simulation started. Press Ctrl and C together to stop.")
11
12 try:
13     while True:
14         # Red light
15         red_led.on()
16         yellow_led.off()
17         green_led.off()
18         print("Red light")
19         sleep(15) # wait 5 seconds
20
21         # Green light
22         red_led.off()
23         yellow_led.off()
24         green_led.on()
25         print("Green light")
26         sleep(30) # wait 5 seconds
27
28         # Yellow light
29         red_led.off()
30         yellow_led.on()
31         green_led.off()
32         print("Yellow light")
33         sleep(5) # wait 2 seconds
34
35 except KeyboardInterrupt:
36     print("\nSimulation stopped by user.")
37
38 finally:
39     # Turn off all LEDs
40     red_led.off()
41     yellow_led.off()
42     green_led.off()
43     print("All LEDs turned off.")
```